

SLF Lure Pitch

Team: Team Lighthouse **Client(s):** Cornell CALS Extension / E&J Gallo Winery / National Grape

Problem statement

Spotted lantern flies are currently impacting grape harvests massively, almost 300 million dollars lost annually on the grape and wine industry in New York State alone. On top of affecting yield, they also impact quality of the harvests, over 10% quality reduction in just 4 years.

Impact

If we can limit their destruction, we can save farmers thousands of dollars and reduce food waste.

Proposed direction: Decoy Lure

A humidifier that will release attractive fumes and poisonous fumes/insecticides at a to be determined rate. There will be a form of sticky tearable tape nearby to trap the SLF upon landing. We will also be able to incorporate speakers easily that emits 60Hz sounds to increase effectiveness. - Humidifier hung around tree(Tree of heaven) or telephone. - Layer of sticky tearable tape wrapped around tree or telephone pole within the region of the humidifier. - A person tears off one layer after a set period of time.

Why it's better?

- Will not affect the harvests, and pulls SLF away from grape vines
- Reduces population of SLF.
- Easily Scalable

End-of-semester proof-of-concept: A humidifier with an adapter that can be attached to a tree or pole.

Key risks / unknowns

- Requires more manpower to kill when they're trapped and need to be extracted. This may increase costs for the company. We can test it by placing the trap near a near grapes, monitor and attempt extraction when required.
- Effectiveness vs grapes. If the chemical used are less attractive than grapes, lure is useless. Develop different chemicals to place inside the humidifiers and test effectiveness of lure.
- Catching other bugs/animals, may affect surrounding ecosystem negatively. Place the trap and monitor what other species are attracted by the lure.

Questions for the client

1. How would you rank the effectiveness of each luring factor we have in our idea? The choice of chemicals/speaker to be incorporated in the lure.
2. Would it be harmful to the grape vines to plant Trees of Heaven in a nearby vicinity? Where to attach the humidifiers.
3. What are different bugs that may be affected by catching spotted lanternflies and what are things we can do to prevent this? Our choice in lure composition will have to change.

References

- <Source 1 (<https://www.timesunion.com/hudsonvalley/outdoors/articlespotted-lanternflies-upstate-new-york-20822211.php>)>
- <Source 2 (<https://academic.oup.com/jee/article/115/6/2116/6777183>)>
- <Source 3 (<https://academic.oup.com/ee/article/49/5/1049/5900068>)>
- <Source 4 (<https://www.usda.gov/about-usda/news/blog/spotted-lanternfly-reveals-potential-weakness#:~:text=%E2%80%9CThere%20were%20rumors%20that%20lanternflies,of%20%5Bthe%5D%20vibr>)>